Lec3

# Vector Spaces and Subspaces

CS 556
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# Vector Spaces and Subspaces



# Vector Spaces

- The space $\mathbb{R}^n$ consists of all columns vectors **v** with $n$ components.

- We can add any two vectors in $\mathbb{R}^n$, and we can multiply any vector **v** by any scalar $c$.

- Vector spaces must be closed under addition and multiplication

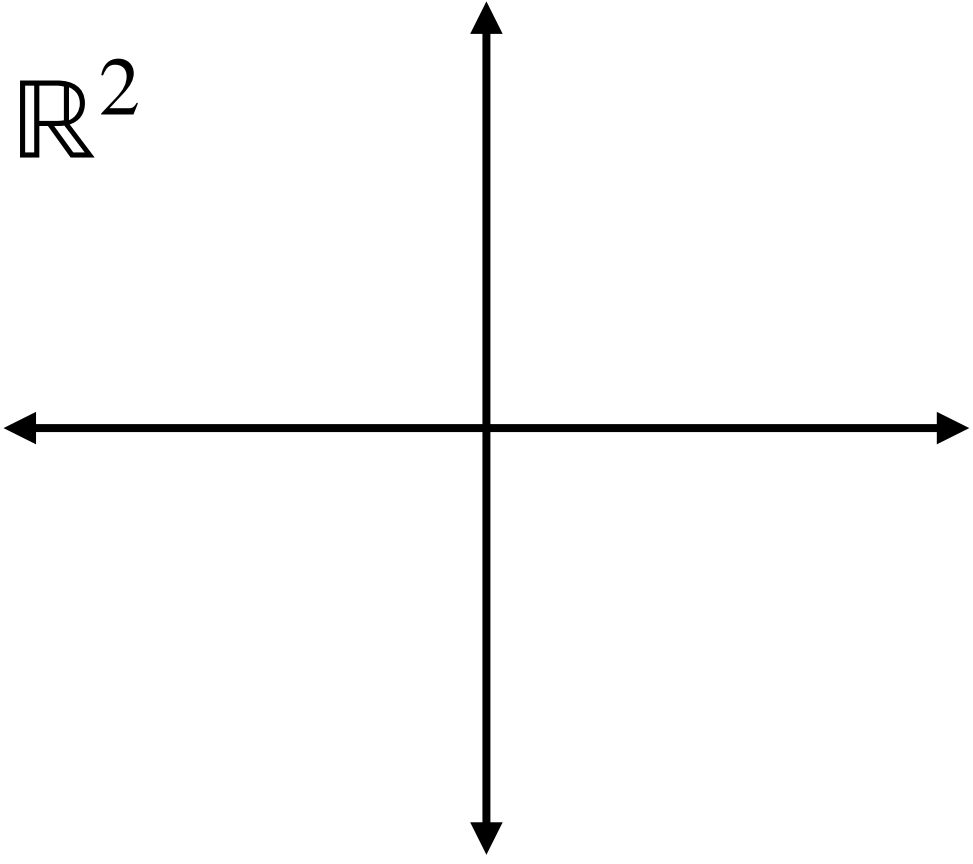
Examples:

- The vector space $\mathbb{R}^2$ is represented by the $xy$ plane. Vector examples in this space: (3, 2), (0, 0) etc.

- The vectors space $\mathbb{R}^3$ is represented by the $xyz$ 3-dimensional space. Vector examples in this space: (1, 2, 3), (0, 0, 0) etc.

# Vector Spaces

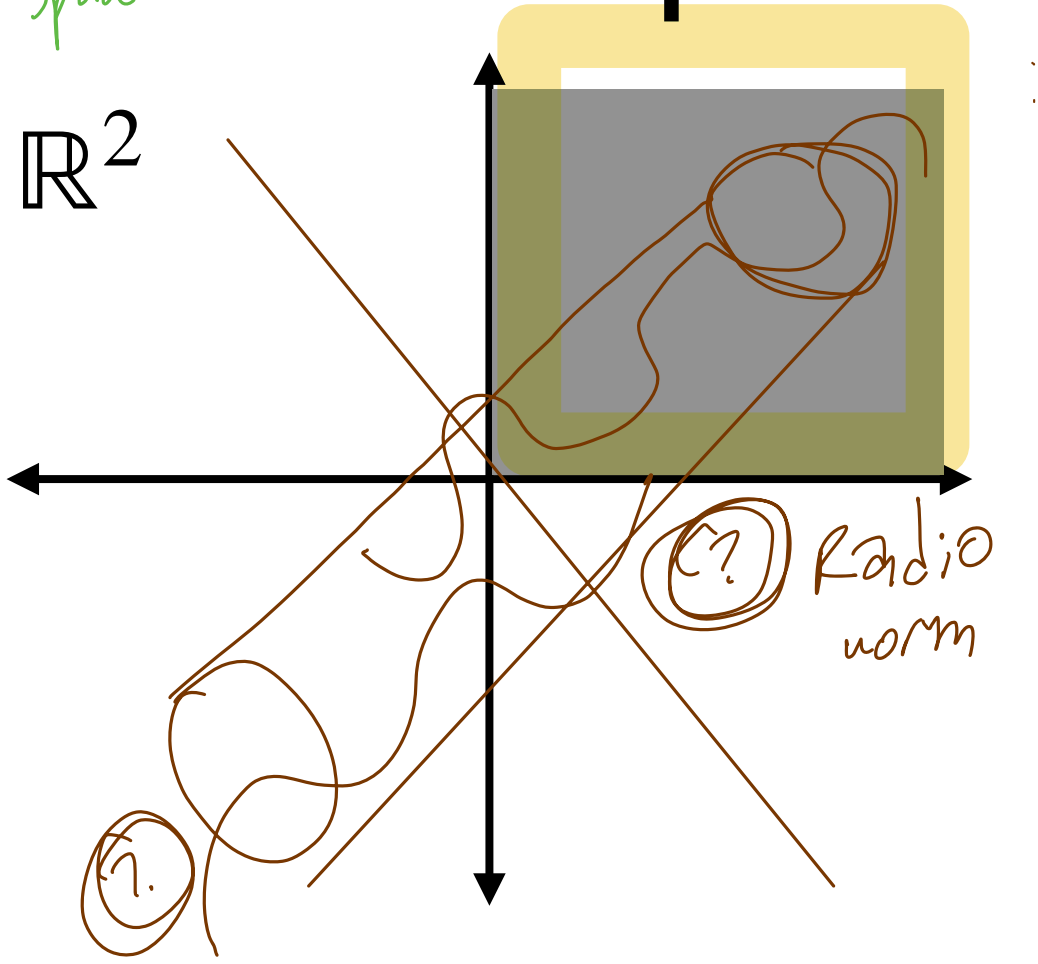
$\mathbb{R}^2$



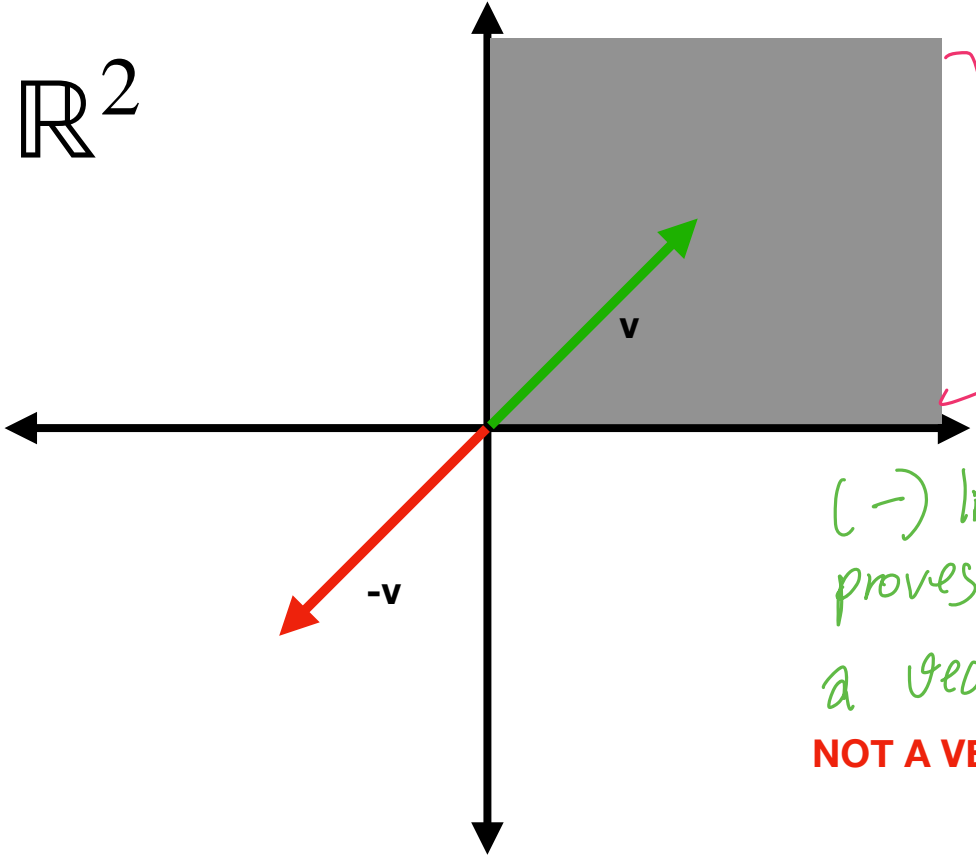
(?)

Is this a vector space?

# Vector Spaces



# Vector Spaces




No!

vector subspace.

(-) linear combination proves it is not a vector space.

**NOT A VECTOR SPACE!!!!**

# 2 Subspace

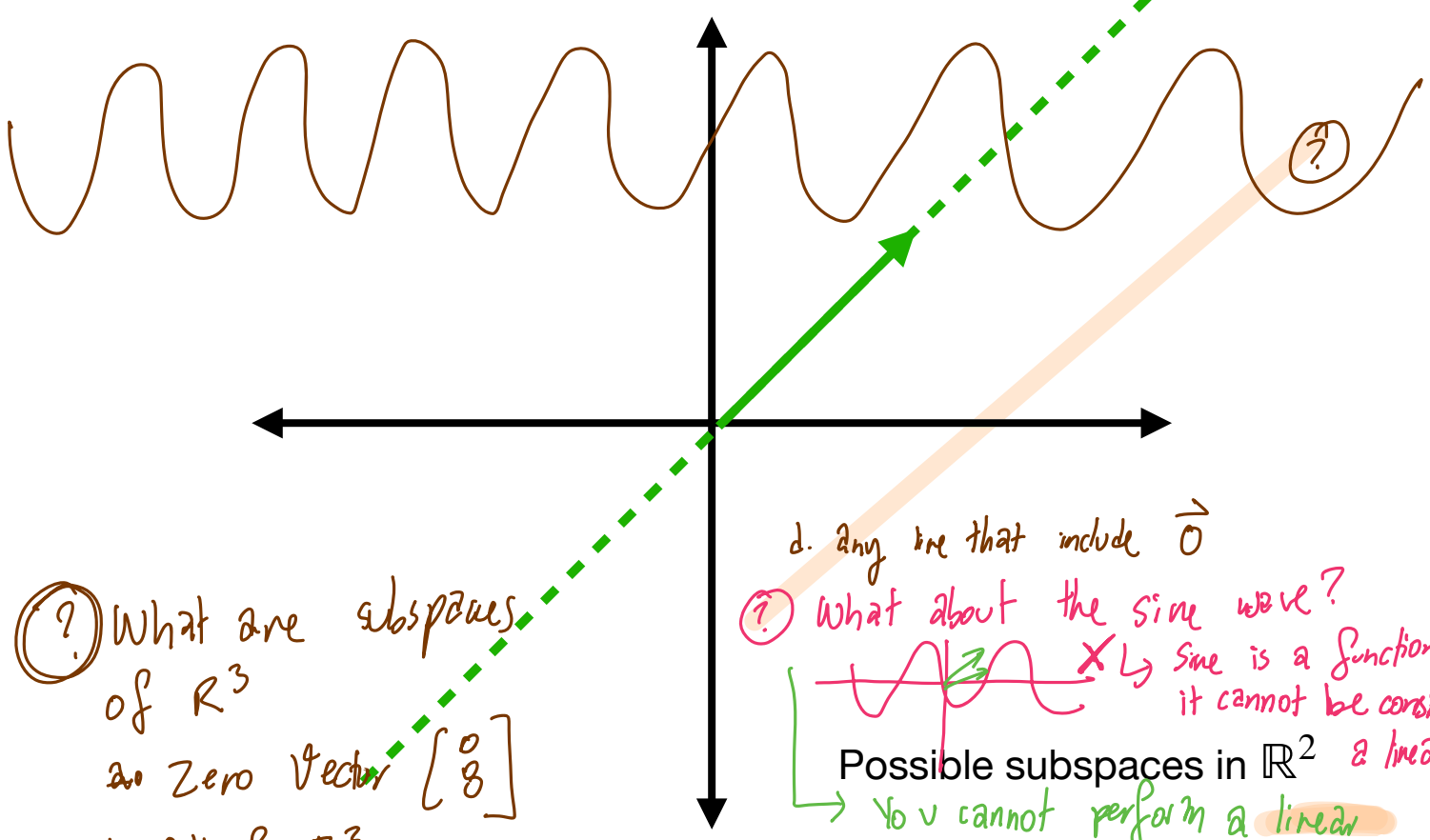
what is the purpose of

A subspace is defined as a set of all vectors that can be created by taking linear combinations of some vectors or a set of vectors.

Formally, a subspace is the set of all vectors that satisfy the following conditions:

- Must be closed under addition and multiplication
- Must contain the zero vector

$$\forall v, w \in \mathbf{V}, \forall \lambda, \mu \in \mathbb{R};\ \lambda\mathbf{v} + \mu\mathbf{w} \in \mathbf{V}$$



? What are subspaces of $R^3$

a. Zero Vector $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$

b. All of $R^3$

? Is $R^2$ a subspace of $R^3$?

X ↳ NO! it must have a 3rd dimension.

c. Any plane that include $\vec{0}$

d. any line that include $\vec{0}$

? What about the sine wave?



X ↳ Sine is a function, it cannot be considered a linear vector.

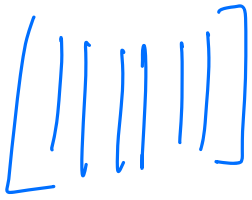
↳ You cannot perform a linear combination on a vector to obtain a wave.

Possible subspaces in $\mathbb{R}^2$

- All of $\mathbb{R}^2$
- Lines that pass through the origin
- The zero vector $\mathbf{z} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$



# Column Space

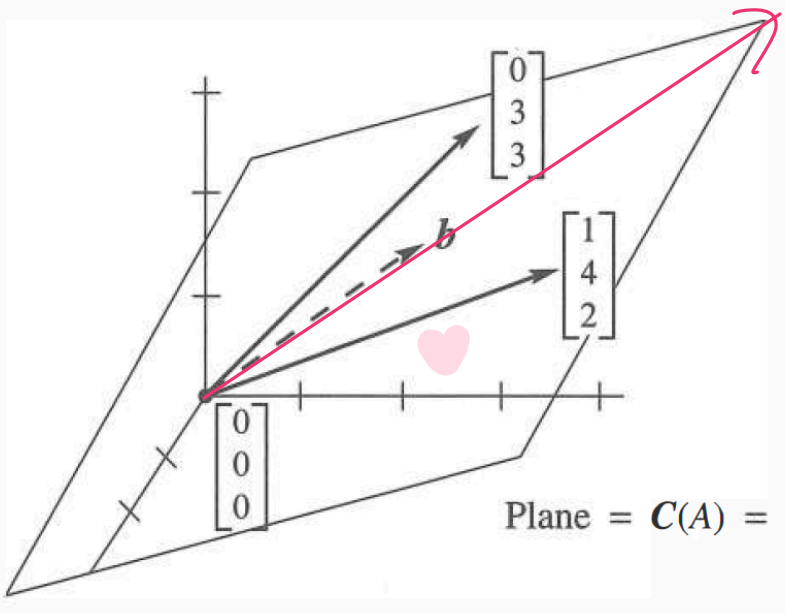


- The column space of a matrix $A$ consists of all the linear combinations of the columns of $A$.
- The combinations are all possible vectors $A\mathbf{x}$, which fill the columns space denoted by $C(A)$.
- The system $A\mathbf{x} = b$ is solvable if and only if $b$ is in the columns space of $A$.

if b is part of the column space, the system is solvable.

# Column Space Example

1. Linear combinations are all in the column space of A.




$$A = \begin{bmatrix} 1 & 0 \\ 4 & 3 \\ 2 & 3 \end{bmatrix}$$

$$\boldsymbol{b} = .4 \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} + .3 \begin{bmatrix} 0 \\ 3 \\ 3 \end{bmatrix}$$

$$A\boldsymbol{x} = \boldsymbol{b} \text{ has } \boldsymbol{x} = \begin{bmatrix} .4 \\ .3 \end{bmatrix}$$

Plane = $\boldsymbol{C}(A)$ = all vectors $A\boldsymbol{x}$

**Credit: Intro to Linear Algebra. Strang Book**

# 4 Null Space

- The null space of a matrix $A_{mxn}$ consists of all the solutions to $A\mathbf{x} = 0.$ – set of vectors satisfying this criteria.
- The solution vectors $x$ have $n$ components. They are vectors in $\mathbb{R}^n$, so the null space is a subspace of $\mathbb{R}^n$. The columns space $C(A)$ is a subspace of $\mathbb{R}^m$.

Q1: Compute the Null space

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \end{bmatrix}$$

$$\begin{matrix} \\ R_2 - 2R_1 \\ R_3 - 3R_1 \end{matrix} \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{matrix} R_1 - R_2 \\ -1R_2 \\ R_3 + 2R_2 \end{matrix} \begin{bmatrix} 1 & 1 & 2 \\ 0 & -1 & -1 \\ 0 & -2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

**REDUCED ROW-ECHELON FORM**

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x + 1 = 0$$
$$y + 1 = 0$$

**Pivot Columns** **Free Column**

no pivot!

$$\mathbf{s} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

**Special Solution**

$$\mathbf{Z} = c \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

**Null Space**

? How can we relax constraints or widen the range of a specific subspace?

Any linear combination of this

5

? What does column space look like in crochet?

# Four Fundamental Subspaces

? What does column space look like for hair?

The four fundamental subspaces of $A_{mxn}$.

| Name | Notation | Note |
|---|---|---|
| Column Space | $C(A) \in \mathbb{R}^m$ | All combinations of the columns of matrix A. |
| Null Space | $N(A) \in \mathbb{R}^n$ | |
| Row Space | $C(A^T) \in \mathbb{R}^n$ | All combinations of the rows of matrix A. |
| Left Null Space | $N(A^T) \in \mathbb{R}^m$ | |

become row space.




# Complete Solution System of Linear Equations

# 2.1 Complete Solution to Ax = b

- Set all free variables to 0, then solve $Ax = b$ for pivot variables to find $x_{particular}$ .
- Find the null space: $x_{null\ space}$ .
- The compete solution is:

$$x = x_{particular} + x_{null\ space}$$

$$\begin{bmatrix}1&1&3\\1&2&4\\2&2&6\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}15\\21\\30\end{bmatrix}$$

# Q1

① Find $x_p$ particular

$$\left[\begin{array}{ccc|c}1&1&3&15\\1&2&4&21\\2&2&6&30\end{array}\right] \xrightarrow{\begin{array}{l}R2=R2-R1\\R3=R3-2R1\end{array}} \left[\begin{array}{ccc|c}1&0&2&9\\0&1&1&6\\0&0&0&0\end{array}\right] = \begin{bmatrix}9\\6\\0\end{bmatrix}$$

Free var $z$

$z=0$

$y=6$

$x=9$

$$x_p=\begin{bmatrix}9\\6\\0\end{bmatrix}$$

② Find $x_n$ null space.

$$\begin{bmatrix}1&0&2\\0&1&1\\0&0&0\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$$

$z=1$

$y=-1$

$x=-2$

$$S=\begin{bmatrix}-2\\-1\\1\end{bmatrix}$$

Complete Solution:

$$x=\begin{bmatrix}9\\6\\0\end{bmatrix}+c\begin{bmatrix}-2\\-1\\1\end{bmatrix},\ c\in\mathbb{R}$$

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# Q2

$$\begin{aligned}a+2b+3c+5d&=0\\2a+4b+8c+12d&=6\\3a+6b+7c+13d&=-6\end{aligned}$$

$2\ 4\ 8\ 12\ 6$ and $3\ 6\ 7\ 13\ -6$

$-2\ -4\ -6\ -16\ 0$ and $-3\ -6\ -9\ -15\ 0$

$-2\ -2$

① $$\left[\begin{array}{cccc|c}1&2&3&5&0\\2&4&8&12&6\\3&6&7&13&-6\end{array}\right] \xrightarrow{\begin{array}{l}R_2=R_2-2R_1\\R_3=R_3-3R_1\end{array}} \left[\begin{array}{cccc|c}1&2&3&5&0\\0&0&2&2&6\\0&0&-2&-2&-6\end{array}\right] \left[\begin{array}{cccc|c}1&2&3&5&0\\0&0&1&1&3\\0&0&0&0&0\end{array}\right]$$

$x=-1$

$y=2$

$z=3$

$a=0$

$$\begin{bmatrix}-3\\0\\3\\0\end{bmatrix}$$

free variables: $b$, $d$

$b=0, d=0, c=3, a=-3$

② Find the free variables

$x_{nullspace}$: $b=1, d=0, c=0, a=-2$

$$S_1=\begin{bmatrix}-2\\1\\0\\0\end{bmatrix}$$

$b=0, d=1, c=-1, a=-2$

$$S_2=\begin{bmatrix}-2\\0\\-1\\1\end{bmatrix}$$

③ $$x_n=\alpha\begin{bmatrix}-2\\1\\0\\0\end{bmatrix}+\beta\begin{bmatrix}-2\\0\\-1\\1\end{bmatrix}+\begin{bmatrix}-3\\0\\3\\0\end{bmatrix}$$

Infinite # of soltns.

Q3

$$\begin{bmatrix} 1 & 1 & 2 \\ 3 & -1 & 14 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2 \\ 6 \\ 1 \end{bmatrix} \xrightarrow[R_3 = R_3 - R_1]{R_2 = R_2 - 3R_1} \begin{bmatrix} 1 & 1 & 2 \\ 0 & -4 & 8 \\ 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2 \\ 12 \\ 3 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2 \\ -3 \\ 6 \end{bmatrix}$$

No Sol

Inconsistent set

$0x + 0y + 0z = 6$ ??

No solutions.

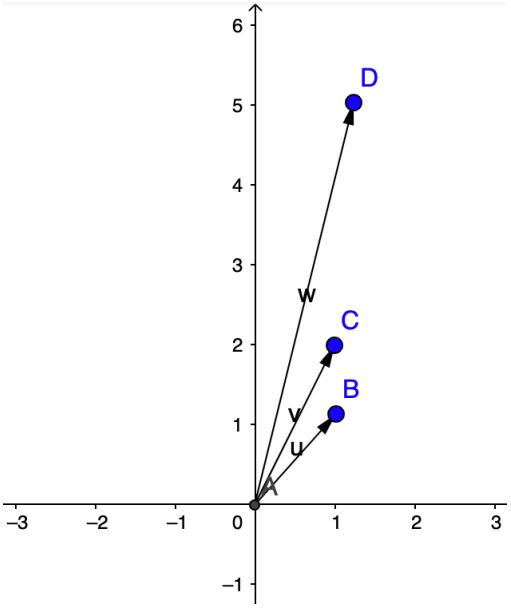
# Matrix Rank

# Matrix Rank

- The rank of a matrix denoted by **r** or **rk(A)** or **rank(A)** is the number of pivot columns.
- Single number the provides insights into the amount of information that is contained in the matrix.
- $r \in \mathbb{N}, s.t.\ 0 \leq r \leq min\{\#cols, \#rows\}$

# Matrix Rank

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 2 & 1 & 0 & 5 \end{bmatrix}$$




Perform Gaussian elimination until the matrix is in row-echelon form and then count the number of pivot columns

# Computing the rank

$$A = \begin{bmatrix} 1 & 2 & 1 \\ -2 & -3 & 1 \\ 3 & 5 & 0 \end{bmatrix} \qquad rk(A) = ?$$

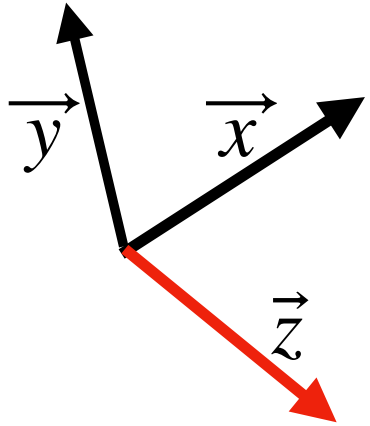
# Linear Independence

# Linear Independence

The sequence of vectors $v_1, \ldots, v_n$ is linearly independent if the only combination that gives the zero vectors is $0v_1 + 0v_2 + \ldots + 0v_n$.

$$x_1v_1 + x_2v_2 + \ldots + x_nv_n = 0$$

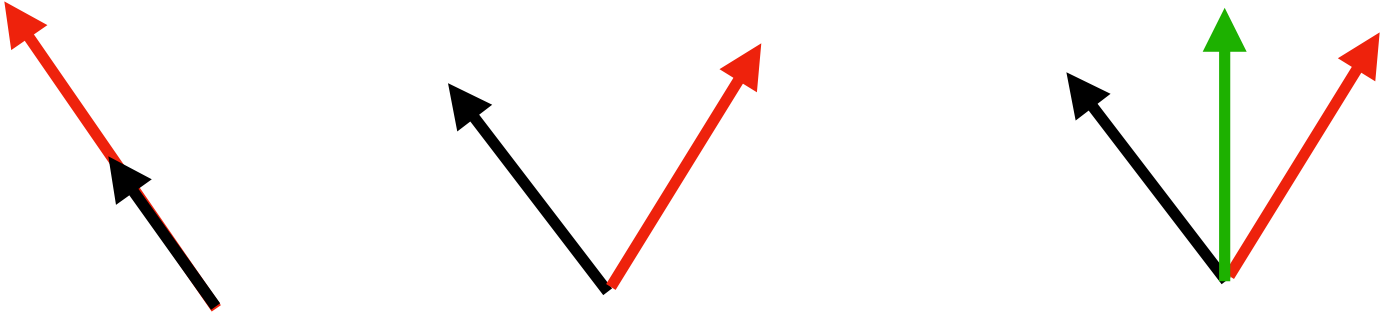
only happens when all x's are zero.



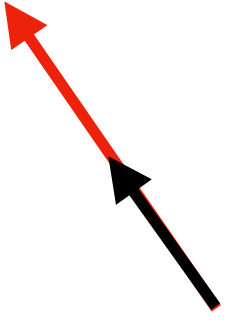

$$z \neq \alpha x + \beta y$$

*z* can not be express as a linear combination of *x* and *y*

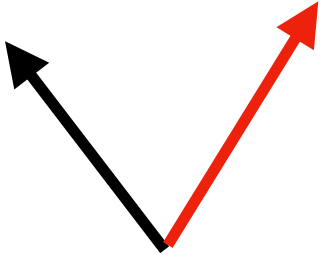
# Are these sets of vectors linearly independent?



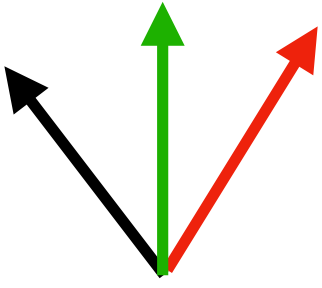
# Are these sets of vectors linearly independent?



NO

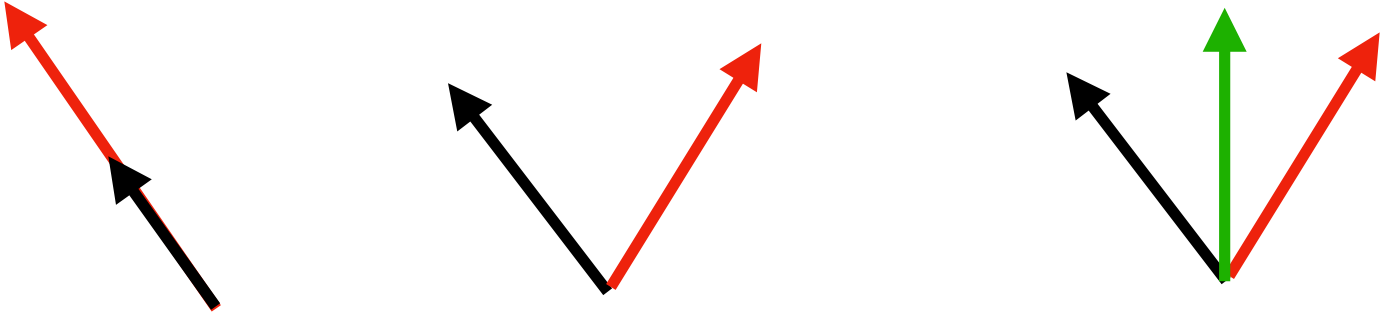


YES



NO

# Are these sets of vectors linearly independent?



**NO** **YES** **NO**

There are a maximum of N independent vectors in $\mathbb{R}^N$.

# Span

Span of a space is defined as all possible linear combinations of all the vectors in that space.

$$span(\{v_1, v_2, \ldots, v_n\}) = \alpha_1 v_1 + \ldots + \alpha_n v_n,\ \alpha_i \in \mathbb{R}$$

# Span Example 1

- $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ span the full space $\mathbb{R}^2$.
- $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$ span the full space $\mathbb{R}^2$.
- $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$ span a line in $\mathbb{R}^2$.

# Span Example 2

To determine if a vector **v** is in the span of a set $S$ we need to check whether **v** can be expressed as a linear combination of vectors in $S$.

$\mathbf{v} = \begin{bmatrix}1\\2\\0\end{bmatrix}$, $\mathbf{S} = \{ \begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\7\\0\end{bmatrix} \}$. Check if $\mathbf{v} \in \mathbf{S}$

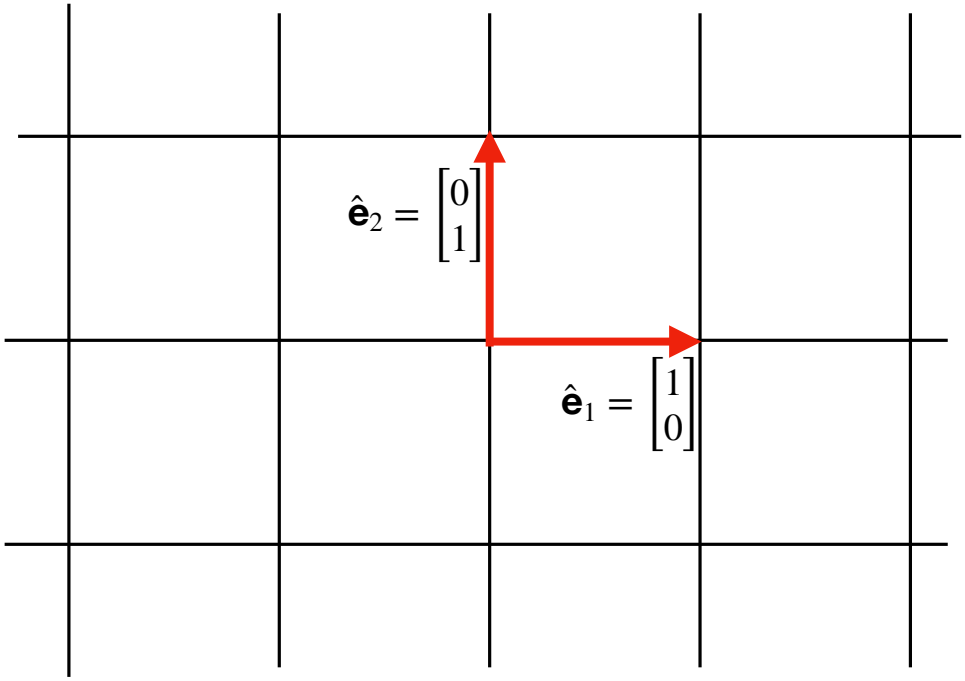
$$\begin{bmatrix}1\\2\\0\end{bmatrix} = \frac{5}{6}\begin{bmatrix}1\\1\\0\end{bmatrix} + \frac{1}{6}\begin{bmatrix}1\\7\\0\end{bmatrix}$$ Yes, $\mathbf{v} \in \mathbf{S}$

# Basis

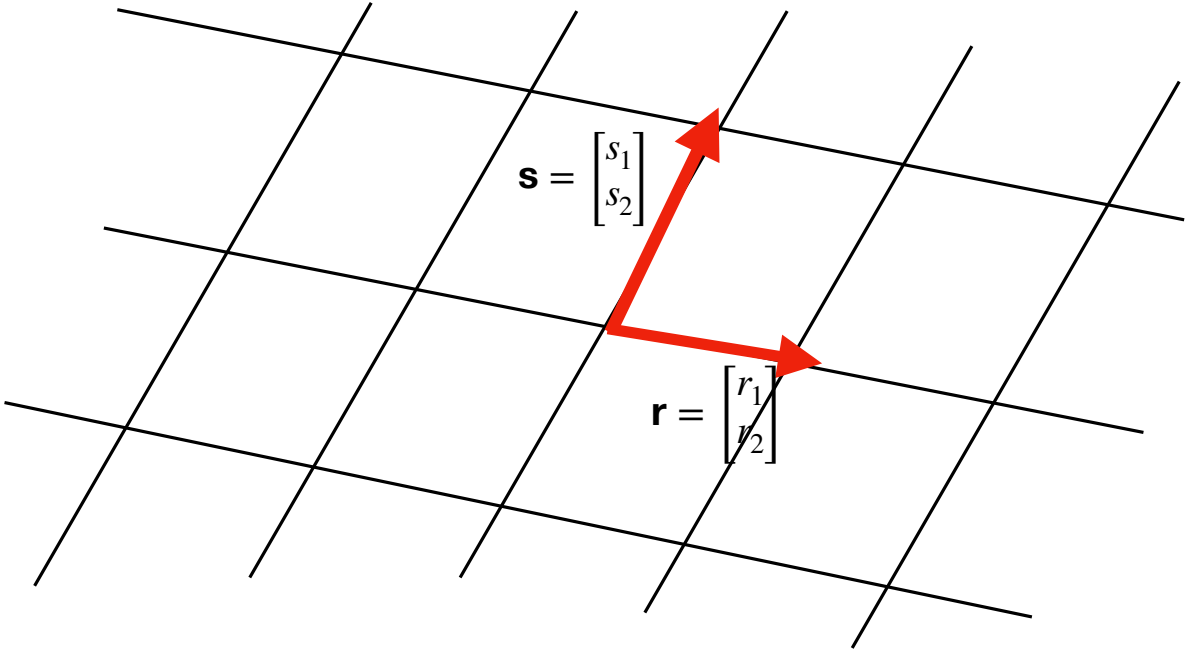
# Basis and Dimension

- A basis for a vector space is a sequence of vectors with two properties:
  - The vectors are linearly independent.
  - The vectors span the space.
- Every vector in the space is a unique combination of the basis vectors.
- A space can be spanned by multiple basis.
- The dimension of a space is the number of vectors in every basis.

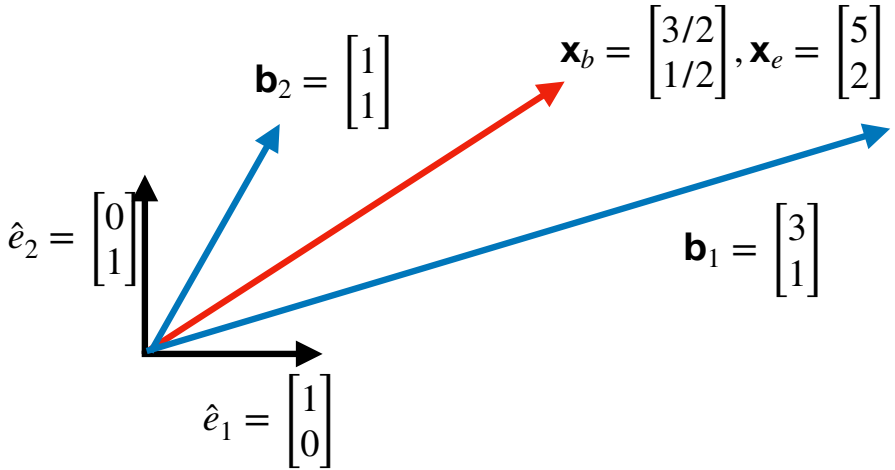
# Natural Basis



# Another Basis



# Changing Basis




$$B = \begin{bmatrix}3 & 1\\1 & 1\end{bmatrix} \qquad \mathbf{x}_b = \begin{bmatrix}3/2\\1/2\end{bmatrix} \qquad \mathbf{x}_e = \begin{bmatrix}3 & 1\\1 & 1\end{bmatrix}\begin{bmatrix}3/2\\1/2\end{bmatrix} = \begin{bmatrix}9/2 + 1/2\\3/2 + 1/2\end{bmatrix} = \begin{bmatrix}5\\2\end{bmatrix}$$

$$B = \begin{bmatrix}3 & 1\\1 & 1\end{bmatrix} \qquad \mathbf{x}_e = \begin{bmatrix}5\\2\end{bmatrix} \qquad \mathbf{x}_b = \begin{bmatrix}3 & 1\\1 & 1\end{bmatrix}^{-1}\begin{bmatrix}5\\2\end{bmatrix} = \frac{1}{2}\begin{bmatrix}1 & -1\\-1 & 3\end{bmatrix}\begin{bmatrix}5\\2\end{bmatrix} = \begin{bmatrix}3/2\\1/2\end{bmatrix}$$

Thank you!